Faculty of Engineering and Technology

Department of Computer Science

Business Intelligence - COMP438

Project Report No.1

Customer Behavior Prediction using Classification and Regression Algorithms in R

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Abstract

The study examines the ability to predict whether or not a customer will make a purchase from their browsing patterns and activity within a session using machine-learning methods. To do so, the researchers used the Online Shoppers Purchase Intentions Data Set from the UCI repository and implemented two classification models: Logistic Regression and Decision Tree.

The data set includes a variety of behavioral, interactional and technical characteristics that represent how users navigate within an online store. These models were created from training and testing performed with established standard evaluation metrics in order to achieve an accurate predictive ability of distinguishing between purchasers and non-purchasers. The researchers also provide insights into the advantages and disadvantages that each model provides in allowing e-commerce businesses to utilize these predictive techniques for improving the decision-making process, personalizing the content of their websites and developing targeted marketing campaigns.

Table of Contents

1. Introduction 6

2. Algorithms 7

2.1 Logistic Regression 7

2.2 Decision Tree 9

3. Datasets Descriptions. 11

4. Results 12

4.1 Logistic Regression Results 12

4.2 Decision Tree Results 13

4.3 Comparison Between the Two Models 14

4.4 Insights and Interpretation 15

5. Conclusion 16

6. References 17

Table of Figure

Figure 1:Sigmoid function used in Logistic Regression 8
Figure 2:Decision Tree Model for Predicting Online Shoppers' Purchasing Intention 10

Table of Table

Table 1:Confusion Matrix (Logistic Regression) 13
Table 2:Confusion Matrix (Decision Tree)..... 14
Table 3:Comparison Between the Two Models..... 14

1. Introduction

Online shopping has grown rapidly to become one of the most popular ways for consumers to shop for items over the last ten years. The increase in e-commerce sites has led to an increased interest from businesses into understanding how consumers navigate through their sites. When retailers understand how users navigate through their site, which pages they spend the most time on, and what actions they perform while visiting the site, they can create a better shopping experience and increase the likelihood that the consumer will complete a purchase.

Every individual who visits a site creates a record of their activity during their online session. This record includes which pages were viewed, how much time was spent viewing various categories or products, and what actions were taken prior to the final purchase selection. By analyzing these behaviors, retailers can gain insight into how customers think and what may lead them to make a purchase. With machine-learning technology available to assist retailers, retailers can utilize this technology to better understand these behaviors and to predict which customers are likely to finalize a purchase. The goal of this study is to develop models that will enable prediction of the likelihood that a user will purchase something at the end of their shopping experience.

The study used two common classification algorithms, **Logistic Regression and Decision Trees**, which were trained and tested to evaluate how accurately they could differentiate between users who made a purchase and those who did not. The objective of comparing the two algorithms is to illustrate the advantages and disadvantages of each one, as well as to show how e-commerce companies may be able to employ machine learning models to enhance personalization, marketing, and overall user experiences.

Overall, this project provides evidence that data-driven approaches can provide online retailers with the ability to make more informed business decisions and create strategies that will increase engagement from potential customers.

2. Algorithms

In this project, we used two main machine learning algorithms to predict online shopper behavior: **Logistic Regression** and **Decision Tree**.

2.1 Logistic Regression

Logistic Regression is a statistical and machine-learning algorithm commonly used for **binary classification problems**, where the target variable has two possible outcomes (e.g., purchase vs. no purchase) [1]. The method models the relationship between a set of independent predictor variables and the probability of belonging to the positive class. Unlike linear regression, Logistic Regression does not directly predict a continuous value. Instead, it applies the **logistic (sigmoid) function**, which transforms a linear combination of predictors into a probability value bounded between 0 and 1 [1][2].

Mathematically, the model uses the **logit transformation**, which connects the predictor variables to the log-odds of the outcome. This makes the relationship between the predictors and the target mathematically tractable while still capturing nonlinear behavior through the sigmoid curve [1]. To estimate the model's parameters, Logistic Regression commonly uses **Maximum Likelihood Estimation (MLE)**, a statistical method that selects parameter values that maximize the likelihood of observing the given data [1][3].

After estimating the probabilities, a **classification threshold** (typically 0.5) is applied to assign each observation to either the positive or negative class. Logistic Regression is widely used because of its simplicity, computational efficiency, and its ability to provide interpretable results. Its probabilistic output and clear mathematical foundation make it a practical choice for many real-world applications, including predicting online purchasing behavior [2][3].

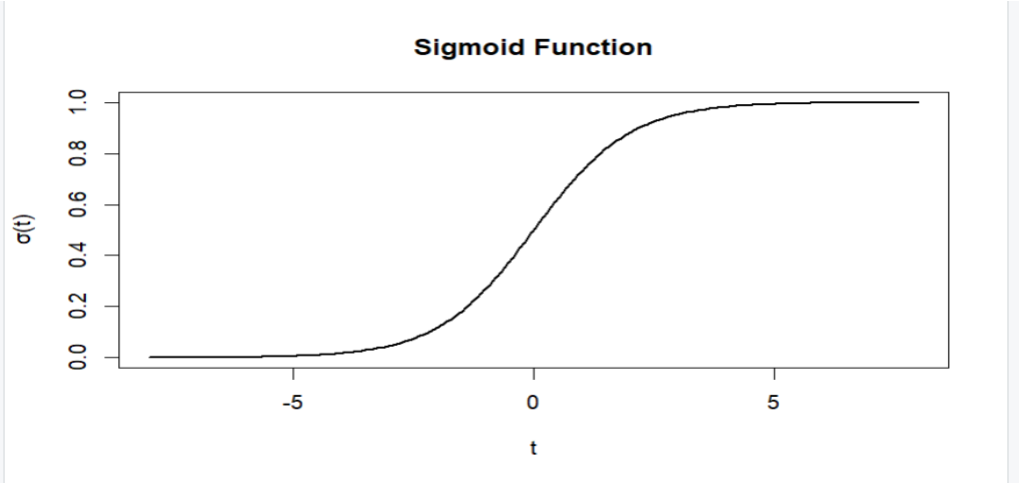

Figure 1:Sigmoid function used in Logistic Regression

The image provides an example of a sigmoid function that is used to convert any value into a probability between "0" (meaning no chance at all) and "1" (meaning certainty). The function's shape is nearly flat at both ends but sharply curves upwards in the middle to create an "S" shape. The sigmoid function is therefore helpful for the model in expressing probabilities associated with predictions.

- For very low input values (that is, the lowest negative numbers), probabilities will be very close to zero; that is to say, there was no chance of a probability being reached. As the input becomes larger (moving toward center), the predicted probability will rise very quickly. Once onto positive numbers (that is, above) the predicted probability will be very close to one (100%).

- The sigmoid function will therefore allow the model to make decisions using probabilities rather than options. Given the shape of the function, input values lead to a more interpretable probability, allowing for easy differentiation between categories, i.e. whether to purchase or not.

2.2 Decision Tree

A **Decision Tree** is a supervised machine-learning method used for both classification and regression tasks. It works by learning a set of simple decision rules from the data and organizing them into a tree-like structure that predicts an outcome based on input features [4]. Each internal node in the tree represents a decision based on a specific feature, each branch represents the result of that decision, and each leaf node represents the final predicted class.

One of the main characteristics of a decision tree is its **interpretability**, as the model visually resembles a flowchart. This allows users to trace how the model arrives at a final decision step-by-step, making it easier to understand compared to many other machine-learning models [4][5]. Decision trees are able to capture **nonlinear relationships**, interactions between variables, and complex patterns in the data without requiring extensive data preparation or assumptions about feature distributions [5].

Decision trees split the dataset into smaller subsets using criteria such as **Gini impurity**, **information gain**, or **entropy**, depending on the chosen algorithm (e.g., CART, ID3, C4.5) [4][5]. At each split, the algorithm selects the feature and threshold that best separate the classes. This process continues recursively until the stopping conditions are met, such as reaching a maximum depth or a minimum number of samples per node.

Although decision trees are powerful and flexible, they can be prone to **overfitting**, especially if the tree becomes too deep or complex. To address this, pruning techniques are applied to remove branches that do not contribute significantly to predictive accuracy, helping improve the model's generalizability [6].

Overall, decision trees are widely used due to their clarity, ease of interpretation, and ability to handle both categorical and numerical features. They are especially valuable in applications where model transparency and interpretability are essential.

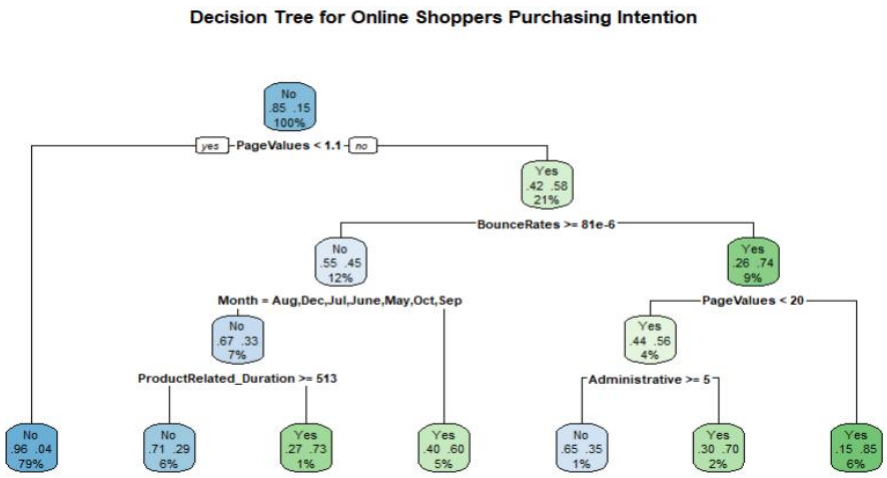

Figure 2:Decision Tree Model for Predicting Online Shoppers' Purchasing Intention

The Decision Tree structure illustrates how the Decision Tree Model, which is trained to determine the likelihood that a Customer will buy online, breaks down the data into smaller clusters of similar values for the various characteristics associated with customer usage.

The root node at the very top of the tree indicates the first and greatest decision made by the model, which in this particular case is the Page Values.

From there, each branch represents some type of inquiry being conducted by the tree (e.g. Page Values<1.1), along with the corresponding conditions (i.e. Bounce Rates>0.000008 or Administration>5).

At the last level of the Decision Tree are the leaf nodes that contain predictions about whether or not a customer is likely to buy with either a "Yes" or "No" response. These leaf nodes also indicate the number of observations that lead to that node and the percentage of "Yes" or "No" responses.

To put it differently, the tree illustrates how the model utilizes characteristics of a user's usage patterns (session duration, page value, bounce rate, administration) in order to separate Users into Potential Buyers or Non-Buyers.

This structured, logical progression allows Users to easily follow the logic of the Decision Tree, allowing for an easy to comprehend understanding of the model's reasoning.

3. Datasets Descriptions.

The dataset collected in this research comes from the UCI Machine Learning Repository and is known as the **Online Shoppers Purchasing Intention Dataset** [7], which contains actual user interactions with an e-commerce website during a one-year period. In total, there are 12,330 records of individual users visiting the site. Each record contains various pieces of data describing how each user interacted with the site, including behavioral, technical and contextual attributes.

The dataset contains both numerical and categorical attributes including the number of pages viewed, total time spent in various areas of the site, probability of exiting or bouncing from the site, and proximity in time to major shopping holidays. Also included are technical attributes (browser type, operating system, traffic source) as well as whether the visit occurred on a weekend.

The revenue variable is being predicted by examining individual user sessions, with the target variable indicating whether or not the user actually made a purchase during the session. Since the number of users who completed a purchase represented less than 15% of the total population represented by this dataset, there is an inherent imbalance within the dataset representative of the way real-world e-commerce businesses operate. The rich combination of behavioral and technical attributes makes this dataset particularly suitable for applying and evaluating classification algorithms such as logistic regression and decision trees.

4. Results

This section presents the experimental results obtained from applying two classification algorithms—**Logistic Regression** and **Decision Tree**—on the *Online Shoppers Purchasing Intention Dataset*.

Both models were evaluated using the same training and testing split (80% training, 20% testing), and their performance was measured using commonly used classification metrics: **accuracy, precision, recall, sensitivity, and specificity**.

4.1 Logistic Regression Results

The Logistic Regression model has an accurate classification rate of 88.36%, suggesting that a majority of the samples were correctly identified by the model. However, the performance of the model with respect to identifying the positive class (i.e. Customers that purchased) was not as good as it could have been.

The following describes the performance of the logistic regression model:

Precision: 0.7327

Recall/Sensitivity: 0.3884

Specificity: 0.9741

The high specificity indicates that the logistic regression model performed well when identifying users without a purchase, but it did not perform well in identifying users who made a purchase which resulted in a low recall rate. When training the model on the dataset, the class "No purchase" was very skewed, hence the performance discrepancies.

Confusion Matrix (Logistic Regression)

Table 1:Confusion Matrix (Logistic Regression)

	Actual No	**Actual Yes**
Predicted No	2030	233
Predicted Yes	54	148

The model produced **233 false negatives**, which explains the low recall. Despite this, the model's precision remained strong, showing that when it predicts "Yes," it is usually correct.

4.2 Decision Tree Results

Compared with the logistic regression method, the decision tree method was the better model and produced an accuracy of 90.10%. Furthermore, it was more effective in detecting positive class predictions than was logistic regression.

Precision: 0.7033

Recall (and Sensitivity): 0.6220

Specificity: 0.9520

The decision tree's increased recall along with balanced performance suggests it had a greater ability to differentiate between customers with a real intent to purchase and those without.

Confusion Matrix (Decision Tree)

Table 2:Confusion Matrix (Decision Tree)

	Actual No	**Actual Yes**
Predicted No	1984	144
Predicted Yes	100	237

The model achieved significantly fewer false negatives (144 instead of 233), demonstrating a more reliable identification of purchasing customers.

4.3 Comparison Between the Two Models

Table 3:Comparison Between the Two Models

Metric	**Logistic Regression**	**Decision Tree**
Accuracy	0.8836	**0.9010**
Precision	**0.7327**	0.7033
Recall	0.3885	**0.6220**
Sensitivity	0.3885	**0.6220**
Specificity	**0.9741**	0.9520

The Decision Tree method showed superiority to the Logistic Regression method in regard to accuracy, recall and sensitivity when correctly identifying users likely to buy.

Conversely, **Logistic Regression** exhibited marginally superior precision and specificity, therefore, Logistic Regression was considered more cautious and less likely to produce false positives.

The nature of the data set was that it was imbalanced; therefore, to some extent, the Logistic Regression model struggled to identify correctly users likely to buy based on their recall score. The Decision Tree model was able to extract better representation of the data by capturing nonlinear relationships in the data set, thus generating a more equitable outcome.

4.4 Insights and Interpretation

1. **Customer purchase intention is not linearly separable**, which explains why decision trees performed better.
   User behavior often depends on interactions between features (e.g., Page Values × Bounce Rates), which trees capture naturally.
2. **Logistic Regression remains useful for interpretability**, but it oversimplifies the decision boundaries, leading to poor detection of "Yes" cases.
3. **The Decision Tree offers actionable insighats**
   The tree reveals important decision rules, such as:
   - High *Page Values* greatly increase purchase probability.
   - High *Bounce Rates* significantly reduce the chance of purchase.
   - Certain months or browsing sessions correspond with higher buying intent.
4. **For business purposes**, the Decision Tree model is more suitable since it better identifies potential buyers, which is crucial for marketing and targeted recommendations.

5. Conclusion

In this project, machine learning techniques were evaluated to ascertain if they could be employed to forecast if an online consumer will complete a purchase based on the user's browsing history. The project developed and assessed two models to classify the Online Shoppers Purchasing Intention Dataset using the same features. The two models developed were Logistic Regression and a Decision Tree classifier. Both models used the same features and both classifiers were assessed by industry-standard performance metrics, such as accuracy, precision, recall, sensitivity, and specificity.

The results indicated that **Decision Tree** classifier was superior in performance overall. It achieved a higher accuracy rating along with significantly higher recall and sensitivity, meaning that it was better able to identify users that were genuinely interested in making a purchase. Logistic Regression performed reasonably well and provided good specificity while it was less capable of identifying positive (purchasing) cases and therefore not as well suited for this type of task.

The Decision Tree, on the other hand, provided more easily interpreted results while capturing nonlinear relationship(s) in user browsing behaviors so that accurate predictions could be made. This emphasized the advantages of employing flexible, tree-structure-based models to analyze consumer browsing activities within an e-commerce environment.

6. References

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